

CBSE Class 10 Mathematics — Circles

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains *questions. All questions are compulsory.*
1. Questions are grouped into sections A through E.
2. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the center O at a point Q so that $OQ = 13$ cm. The length of PQ is:
(A) (A) 10 cm
(B) (B) 11 cm
(C) (C) 12 cm
(D) (D) 15 cm
2. If tangents PA and PB from a point P to a circle with center O are inclined to each other at an angle of 80° , then $\angle POA$ is equal to:
(A) (A) 50°
(B) (B) 60°
(C) (C) 70°
(D) (D) 80°
3. The number of tangents that can be drawn from a point lying inside a circle is:
(A) (A) 0
(B) (B) 1
(C) (C) 2
(D) (D) Infinite
4. Two concentric circles are of radii 5 cm and 3 cm. The length of the chord of the larger circle which touches the smaller circle is:
(A) (A) 6 cm
(B) (B) 8 cm
(C) (C) 10 cm
(D) (D) 7 cm

5. If TP and TQ are two tangents to a circle with center O so that $\angle POQ = 110^\circ$, then $\angle PTQ$ is equal to:
- (A) 60°
 (B) 80°
 (C) 70°
 (D) 90°
6. Assertion (A): The length of a tangent drawn from an external point to a circle is always equal to the radius of the circle. Reason (R): Tangents drawn from an external point to a circle are equal in length.
- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) Assertion (A) is true but Reason (R) is false
 (D) Assertion (A) is false but Reason (R) is true
7. Assertion (A): If the angle between two tangents drawn from a point P to a circle of radius r and center O is 60° , then $OP = 2r$. Reason (R): The line segment joining the center to the external point bisects the angle between the two tangents.
- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) Assertion (A) is true but Reason (R) is false
 (D) Assertion (A) is false but Reason (R) is true
8. Assertion (A): A tangent to a circle is perpendicular to the radius at the point of contact. Reason (R): The perpendicular distance from the center of the circle to the tangent is equal to the radius of the circle.
- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) Assertion (A) is true but Reason (R) is false
 (D) Assertion (A) is false but Reason (R) is true
9. Assertion (A): The number of tangents that can be drawn from a point inside a circle is zero. Reason (R): A tangent to a circle intersects the circle at only one point.
- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
10. Assertion (A): If the distance between two parallel tangents of a circle of radius 3 cm is 6 cm. Reason (R): Parallel tangents touch the circle at the ends of the diameter.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section C: Short Answer I (3 marks each)

1. If a tangent PA and PB from a point P to a circle with center O are inclined to each other at an angle of 80° , then find $\angle POA$.
2. A quadrilateral $ABCD$ is drawn to circumscribe a circle. If $AB = 6$ cm, $BC = 7$ cm, and $CD = 4$ cm, find the length of AD .
3. From a point Q , the length of the tangent to a circle is 24 cm and the distance of Q from the center is 25 cm. Find the radius of the circle.
4. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.
5. If the angle between two tangents drawn from an external point P to a circle of radius r and center O is 60° , find the length of OP .

Section D: Long Answer (4-5 marks each)

1. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.
2. A quadrilateral $ABCD$ is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$.
3. If a tangent PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80° , find $\angle POA$.
4. Prove that the parallelogram circumscribing a circle is a rhombus.
5. A point P is 13 cm from the centre of a circle. The length of the tangent drawn from P to the circle is 12 cm. Find the radius of the circle.

6. Prove that the lengths of tangents drawn from an external point to a circle are equal.
7. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.
8. A quadrilateral $ABCD$ is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$.
9. If a hexagon $ABCDEF$ circumscribes a circle, prove that $AB + CD + EF = BC + DE + FA$.